

An inequality's solution set

An inequality can be true for an entire interval of real numbers.

$$x < 3$$

How would we show its solution set on a number line?

Interval endpoints

A bracket includes an endpoint; a parenthesis excludes it.

$[-2, 5)$

Which endpoint belongs to this interval?

An unbounded interval

$$x \geq 4$$

How do we write this set in interval notation?

Infinity is not an endpoint value

$(-\infty, 6]$

Why does infinity always use a parenthesis?

Multiply by a negative number

$$2 < 5$$

What inequality results when both sides are multiplied by -3?

Solve a linear inequality

$$4x - 7 \leq 13$$

What is the solution set in interval notation?

Reverse the inequality

$$5 - 3x > 14$$

What is the solution set in interval notation?

Distribution and comparison

$$2(x + 4) \leq 5x - 1$$

What is the solution set?

A compound inequality

$$-4 < 2x + 2 \leq 10$$

What interval satisfies both comparisons?

Intersection means both

$$x > -1 \text{ and } x \leq 4$$

What interval is the intersection?

Union means either

$$x < -2 \text{ or } x \geq 3$$

What is the solution set in interval notation?

All real numbers

$$3(x + 1) < 3x + 5$$

Which real numbers satisfy this inequality?

No real numbers

$$2x + 6 \leq 2x + 1$$

Which real numbers satisfy this inequality?

Absolute value and distance

$$|x - 2| < 3$$

Which points lie less than 3 units from 2?

Include the boundary

$$|x + 4| \leq 2$$

What interval includes all the solutions?

Outside an interval

$$|x - 1| > 5$$

What two intervals make up the solution set?

Isolate the absolute value

$$2|3x - 1| + 4 \leq 12$$

What is the solution set?

A negative bound

$$|x + 1| < -2$$

What is the solution set?

A nonpositive bound

$$|2x - 3| \leq 0$$

Which values satisfy this inequality?

An always-true comparison

$$|x - 7| > -1$$

Which real numbers satisfy this inequality?

A tolerance interval

A rod's length L must differ from 50 centimeters by no more than 0.2 centimeters.

What absolute value inequality expresses the requirement?

A radical's allowed inputs

$$\sqrt{8 - 2x}$$

What inequality describes all real inputs for this expression?